

GOETHE MBA | STUDY SCRIPT

Corporate Finance

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1 Abbreviations and Symbols

1.1 Abbreviations

bps Basis Points
100 bps = 1 %, 1 bp = 0.01 %

FV Future Value
Amount into which today's sum grows by a future date.

IRR Internal Rate of Return
Discount rate that makes the net present value of a project equal to zero.

NPV Net Present Value
Present value of all project cash inflows and outflows, including the initial investment.

PI Profitability Index
Present value of future cash inflows divided by the original project cost in the standard single-upfront-outflow case.

PV Present Value
Value today of payments made at future dates.

RR Rate of Return
Generic classroom shorthand for an investment return or yield.

1.2 Mathematical Symbols

Symbol	Definition
t	Payment date or period index. $t = 0$ represents today.
T	Final period or investment maturity.
C_t	Net cash flow at period t. C_0 often represents an initial negative investment.
C	Constant periodic payment or coupon, depending on context.
C_1	First payment at time 1. Important for growing-perpetuity and growing-annuity formulas.
r	Relevant discount rate or required rate of return per period.
r^*	Internal rate of return. A root of NPV(r), where net present value equals zero.
r_f	Risk-free benchmark rate for the relevant currency and horizon.
r_E, r_D	Required returns on equity and debt, respectively.
w_E, w_D	Proportions of equity and debt in financing. $w_E + w_D = 1$.
g	Constant per-period cash-flow growth rate.
F	Bond face value. Principal paid at maturity.
P_0	Present market price or value of an asset at time $t = 0$.
D_0, D_1	Current or most recent dividend and next-period dividend, respectively.
E_1	Next-period accounting earnings. Per share when compared with a share price.
π	Inflation rate over the relevant period. Expected inflation in an ex-ante real-rate comparison.
r_{nom}, r_{real}	Nominal and real interest rates, respectively.

Symbol	Definition
DF_t	Discount factor for a payment at time t. For a constant discount rate: $(1+r)^{-t}$.

2 Introduction and Financial Value

2.1 What is a company, financial claim, or investment project worth today?

Corporate finance examines how firms raise capital, invest it, and create value. A corporation receives funds from debt and equity investors and invests them in projects. These projects generate future cash flows, which can be used to repay creditors, distribute profits to shareholders, or finance further investments.

Corporate finance therefore addresses three connected decisions:

- **Investment**: Which projects create value?
- **Financing**: How should investments be funded through debt and equity?
- **Conflicts of interest**: How do the different interests of managers and investors influence these decisions?

To evaluate these decisions, finance combines economics (choices and prices), accounting (financial information), and statistics (data and uncertainty). The same valuation principles apply across financial markets, including stocks, bonds, foreign exchange, and derivatives.

2.2 What Determines Financial Value?

The value of a financial claim depends primarily on two characteristics of its future cash flows:

- **Time**: When will the money be received? Earlier payments are generally more valuable.
- **Risk**: How certain is the payment? Safer payments are generally more valuable than equally large risky payments.

Time value of money: Receiving money today allows immediate consumption or investment. Saving means giving up consumption today in exchange for future consumption. The market interest rate compensates investors for waiting and represents the opportunity cost of capital. Consequently, EUR 100 today is worth more than EUR 100 received in one year when the relevant interest rate is positive.

Cash flows occurring at different times must be converted into values at a common date, usually today ($t = 0$), using an appropriate discount rate. The valuation of an investment requires comparing its initial cost with the present value of its future cash flows, accounting for the timing and risk of those payments.

3 Valuation of Safe Cash Flows

3.1 Returns, Future Value and Present Value

Investing money means giving up money today in exchange for a future payment. The return measures how much the investor earns.

Example

Assume an investor invests 100 € today and receives 105 € after one year.

- **Payoff**: The total amount received from the investment: 105 €.
- **Net return**: The amount earned after subtracting the initial investment:

$$105 - 100 = 5\,€.$$

- **Rate of Return (RR) (yield)**: The net return expressed as a percentage of the initial investment:

$$r = \frac{105 - 100}{100} = 5\%.$$

While the RR measures an investment's performance, an interest rate measures the cost of borrowing money. Interest rates and changes in interest rates are often expressed in Basis Points (bps): For example, an increase in the interest rate from 5% to 5.25% equals 25 bps.

The same investment can be viewed from two perspectives:

- **Future Value (FV)**: How much will money invested today be worth at a future date?
- **Present Value (PV)**: How much is a future payment worth today?

For an initial price P_0 and a certain payoff C_1 one period later,

$$r = \frac{C_1 - P_0}{P_0}, \qquad \mathrm{FV}_T = \mathrm{PV}_0(1+r)^T, \qquad \mathrm{PV}_0 = \frac{C_T}{(1+r)^T}. \tag{1}$$

FV compounds an investment forward, PV discounts a future payment back. T is the number of periods, and r is the per-period rate. The discount factor for t periods is $(1+r)^{-t}$.

Example

100 € invested for one year at 5% becomes 105 €. Conversely, 100 € due in one year has PV $= 100/1.05 = 95.24$€ at the same rate. For a fixed future payment, a higher required return reduces its present value. For example, if the required return increases from 5% to 10%, the present value of 105 € falls from 100 € to 95.45 €. This inverse price–yield relationship is especially important for bonds.

3.2 Net Present Value and Investment Prices

An investment project requires spending money today to generate future cash flows. To determine whether it creates value, all cash flows must first be expressed in today's money.

The Net Present Value (NPV) measures how much wealth an investment creates or destroys today:

$$\boxed{\mathrm{NPV}(r) = \sum_{t=0}^{T} \frac{C_t}{(1+r)^t} = C_0 + \sum_{t=1}^{T} \frac{C_t}{(1+r)^t}.} \tag{2}$$

Here, C_t represents the net cash flow in period t, with expenses recorded as negative values. The initial investment C_0 occurs today ($t = 0$) and is therefore not discounted.

Investment decision:

- NPV > 0: Accept. The project creates value.
- NPV $= 0$: The project earns exactly the required return.
- NPV < 0: Reject. The project destroys value.

Example

A project costs 1,000 € today and generates 200 € in year 2 and 900 € in year 3. There is no payment in year 1. The discount rate is 5%.

$$\text{NPV} = -1{,}000 + \frac{200}{1.05^2} + \frac{900}{1.05^3} = -41.14\,€.$$

The future payments are worth only 958.86 € today, less than the 1,000 € investment cost. **Decision**: Reject the project because it reduces present wealth by 41.14 €.

4 Capital-Budgeting Rules

Capital budgeting is the process of deciding which investment projects a company should undertake. While NPV measures the wealth a project creates, other decision rules evaluate investments using percentage returns, profitability ratios or recovery periods. For mutually exclusive projects, where only one alternative can be selected, the objective is to choose the feasible project with the highest NPV.

4.1 Internal Rate of Return

The Internal Rate of Return (IRR) is the discount rate at which a project's NPV equals zero. It represents the project's break-even rate of return, not the monetary value created.

$$\boxed{\text{NPV}(r^*) = \sum_{t=0}^{T} \frac{C_t}{(1+r^*)^t} = 0.} \tag{3}$$

For a conventional investment (initial outflow followed by inflows) with one economically relevant IRR, the decision depends on the required market return r:

$$r^* > r: \text{ Accept;} \qquad r^* = r: \text{ Indifferent;} \qquad r^* < r: \text{ Reject.}$$

Example

The investment project in Section 3.2 has an IRR of approximately 3.44%. Since investors require 5%, the project should be rejected. The IRR and NPV rules therefore lead to the same decision for this conventional investment.

The NPV profile shows how a project's NPV changes with the discount rate. The point where the curve crosses zero is its IRR.

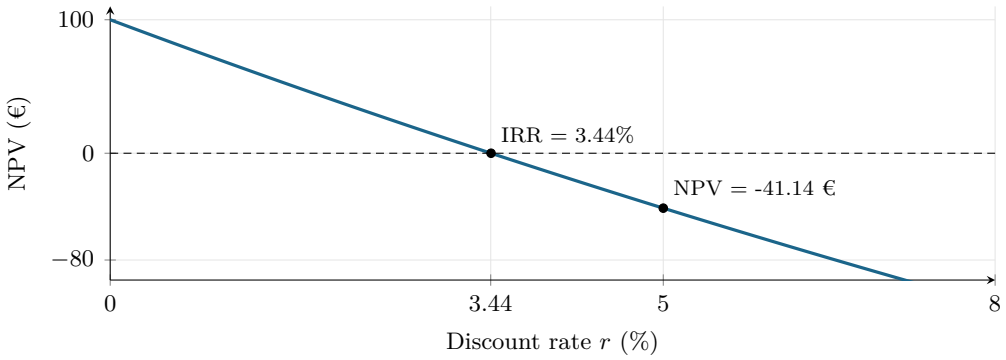


Figure 1: NPV profile: The project breaks even at 3.44% and has a negative NPV at the required return of 5%.

4.2 Limitations of the IRR Rule

IRR is easy to interpret as a percentage, but it can lead to incorrect decisions in three situations:

- **Multiple or no IRRs**: The IRR is found by calculating the discount rate at which NPV equals zero. Depending on the cash flows, this equation can have multiple solutions or none. In the lecture example, an investor pays 100 € today and receives 10 € in year 1 and 110 € in year 2. The equation produces two solutions: 10% and −200%. Only 10% is economically meaningful because the standard discount rate must be greater than −100%. For projects with more complex cash-flow patterns, several valid IRRs may exist, making the investment decision ambiguous.
- **Financing projects**: When money is received initially and repaid later, IRR represents a financing cost rather than an investment return. A higher IRR can therefore be disadvantageous.
- **Different project sizes**: IRR measures percentage returns, not total wealth created. A smaller project can have a higher IRR but generate a lower NPV.

Example

Two mutually exclusive projects have a one-year maturity and a required return of 5%.

Project	Cost	Payoff	IRR	NPV
1	1 €	2 €	100%	0.90 €
2	100 €	150 €	50%	42.86 €

Although Project 1 offers the higher percentage return, Project 2 creates substantially more wealth.

For mutually exclusive projects, NPV takes priority over IRR because it measures the actual value created.

4.3 Profitability Index and Payback Period

Besides NPV and IRR, companies use two other methods to evaluate investments: the Profitability Index (PI), which measures value relative to investment cost, and the payback period, which measures how quickly the initial investment is recovered.

PI measures the present value of future cash inflows per euro invested:

$$\text{PI} = \frac{\text{PV(future cash inflows)}}{\text{initial investment}}.$$

A PI of 1.20 means that every euro invested generates 1.20 € in present value of future cash inflows.

Decision rule: Accept if PI > 1, assuming a single initial outflow followed by future inflows. This corresponds to a positive NPV.

Limitation: The PI measures relative profitability rather than total wealth created. It may therefore favor smaller projects with a higher ratio but a lower NPV. Additional outflows in later periods complicate the rule.

Payback period measures how long it takes for cumulative cash inflows to recover the initial investment.

Decision rule: Accept if the investment is recovered within a predefined maximum payback period.

Limitation: Standard payback ignores the time value of money and cash flows received after recovery.

Example

The lecture compares three projects with the same initial investment of 5 €, but different future cash flows:

Project	Y1	Y2	Y3	Y4	Payback
Project 1	-5 €	8 €	–	–	1 year
Project 2	-5 €	4 €	100 €	–	2 years
Project 3	-5 €	4 €	0 €	100,000 €	3 years

Y1 represents the initial investment. The payback period counts the years needed to recover that cost. The payback rule favors Project 1 because it recovers its cost fastest. However, it ignores the substantial 100,000 € payment Project 3 generates in Y4.

Discounted payback works like the regular payback period, but future cash flows are discounted to their present value before calculating when the initial investment is recovered. For example, an investment of 100 € generates 60 € in year 1 and 50 € in year 2. Regular payback recovers the investment in year 2 ($60 + 50 = 110$ €). At a discount rate of 10%, however, these payments are worth only 95.87 € today. The investment has therefore not yet been recovered in present-value terms. Both payback methods ignore cash flows after the recovery date. In contrast, NPV considers all cash flows and their time value, making it the benchmark for maximizing wealth.

5 Risk-Free Rate, Default Risk and Inflation

The discount rate determines how much future cash flows are worth today. Choosing an appropriate rate requires considering three factors: the risk-free market return, the possibility that payments will not be received, and inflation.

5.1 Risk-Free Benchmark and Treasury Yields

The risk-free rate is the return on an investment assumed to make its promised payments with certainty. It serves as a benchmark for evaluating riskier investments. The benchmark must match the currency and time horizon of the cash flows being valued. For investments in US dollars, the lecture uses US Treasury securities (government debt) as a practical benchmark.

For example, the 10-year constant-maturity Treasury rate represents the market yield associated with US government debt of approximately ten years' maturity. Although repayment is considered very safe, the market price of a Treasury bond can still change when interest rates fluctuate.

For bonds with unchanged promised payments, higher demand tends to increase prices and reduce yields.

Several factors may influence Treasury yields:

- **Investor demand**: Purchases of government bonds can increase prices and reduce yields. The lecture discusses foreign investors, including China, and the demand of ageing investors as possible influences.
- **Inflation expectations**: Higher expected inflation can increase the nominal return investors require.

5.2 Default Risk and Credit Ratings

The risk-free benchmark assumes that promised payments will be received. Corporate debt introduces an additional risk: the borrower may be unable to meet its obligations.

Default risk is the possibility that a borrower fails to pay interest or repay principal as promised. Investors generally require higher promised yields to compensate for greater credit risk.

Credit ratings assess a borrower's creditworthiness using financial information and qualitative judgments. The lecture introduces three major rating agencies: Moody's, Standard & Poor's and Fitch.

Moody's classifies bonds according to their credit risk. The ratings distinguish between investment-grade bonds (lower risk) and speculative-grade bonds (higher risk):

Rating	Credit-risk assessment
Investment Grade	
Aaa	Minimal credit risk; highest quality.
Aa	Very low credit risk.
A	Low credit risk.
Baa	Moderate credit risk; lowest investment-grade category.
Speculative Grade	
Ba	Substantial credit risk.
B	High credit risk.
Caa	Very high credit risk.
Ca	Likely in or near default.
C	Typically in default; little prospect of recovery.

5.3 Inflation and Purchasing Power

Inflation (π) measures the percentage increase in the general price level over a period. A price index tracks the price level, while the inflation rate measures how quickly that index changes. To understand how inflation affects investment returns, distinguish between:

- **Nominal interest rate** (r_{nom}): The stated return on an investment, measured in money, without adjusting for inflation.
- **Real interest rate** (r_{real}): The return after accounting for changes in purchasing power.

The exact relationship between nominal returns, real returns and inflation is:

$$1 + r_{\text{real}} = \frac{1 + r_{\text{nom}}}{1 + \pi}, \qquad r_{\text{nom}} = (1 + r_{\text{real}})(1 + \pi) - 1 \approx r_{\text{real}} + \pi. \tag{4}$$

The approximation is useful when rates are relatively small because it ignores the product $r_{\text{real}}\pi$.

Example

An investment earns a nominal return of 5% over one year, while inflation is 3%.

$$r_{\text{real}} = \frac{1.05}{1.03} - 1 = 1.94\%.$$

Although the investor earns 5% more money, their purchasing power increases by only 1.94%. The simple approximation would give 2%.

When evaluating investments in advance (ex-ante), future inflation is unknown. Therefore, expected inflation is being used. If the required real return stays constant, higher expected inflation leads to a higher required nominal return.

Inflation affects investment valuation in two ways:

- Nominal cash flows: The actual amount of money expected to be received in the future.
- Real cash flows: The future payment expressed in today's purchasing power, after adjusting for inflation.

Discounting rule: Nominal cash flows must be discounted using nominal rates, while real cash flows must be discounted using real rates. Both approaches give the same PV when applied consistently.

Example

An investment pays 103 € in one year. Expected inflation is 3%, and the required real return is 2%.
Due to inflation, the future 103 € has the same purchasing power as 100 € today. The corresponding nominal required return is $1.02(1.03) - 1 = 5.06\%$.
The present value can therefore be calculated in two ways:

$$\text{PV}_{\text{nom}} = \frac{103}{1.0506} = 98.04\,€, \qquad \text{PV}_{\text{real}} = \frac{100}{1.02} = 98.04\,€.$$

Both methods produce the same present value because cash flows and discount rates are consistently adjusted for inflation.